



COMPUTER VISION

Teaching Machines to See and Understand the Visual World

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What Is Computer Vision?

Computer Vision is the field of artificial intelligence that trains computers to **interpret and understand visual information** from images and videos — much like the human eye and brain do.



Captures pixels from a camera or sensor as raw data



Processes that data using algorithms and neural networks



Extracts meaning: objects, faces, motion, and context

HUMAN VISION vs. COMPUTER VISION



Human Vision

Eyes capture light; the brain instantly recognizes objects, depth, and context using decades of learned experience.



Computer Vision

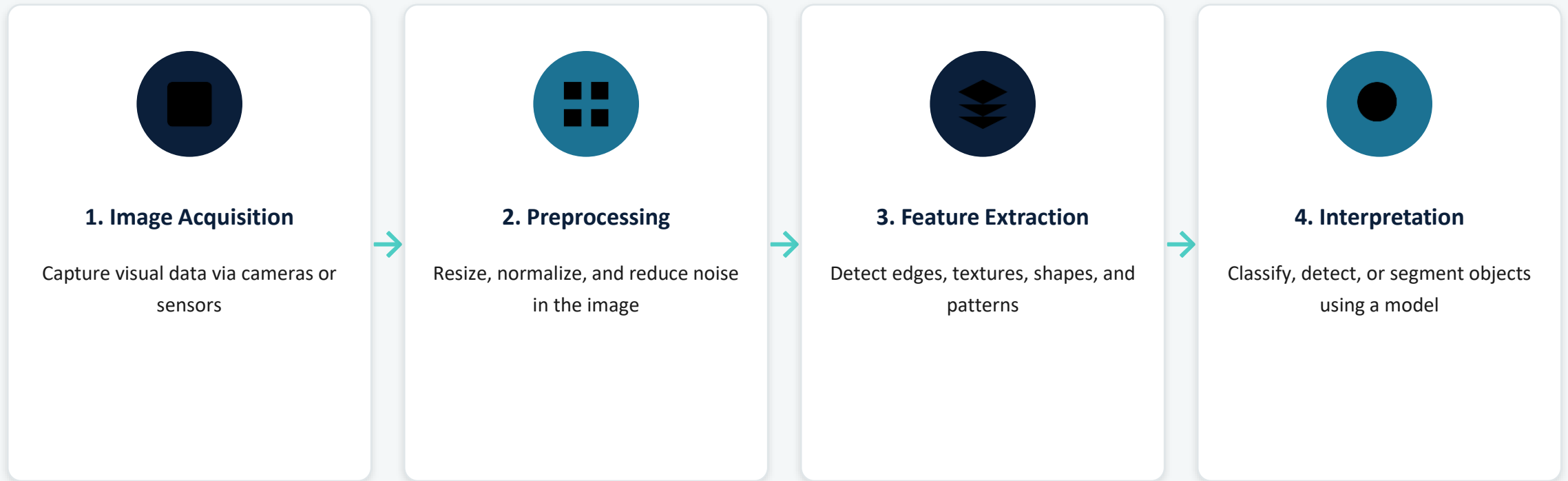
Cameras capture pixel arrays; trained models detect patterns statistically, learning from millions of labeled examples.



Goal: bridge the gap between raw pixels and real understanding

How Computer Vision Works

A typical pipeline from raw image to actionable output



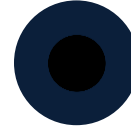
Modern pipelines increasingly use end-to-end deep learning, letting a single neural network learn features and decisions together.

Key Techniques & Concepts



Image Classification

Assigning a single label to an entire image (e.g., "cat" or "dog")



Object Detection

Locating and labeling multiple objects with bounding boxes



Image Segmentation

Classifying every pixel to outline exact object shapes



Convolutional Neural Networks (CNNs)

Deep learning architecture that powers most modern vision models

Real-World Applications

Computer vision already powers everyday technology across industries



Healthcare

Detecting tumors and anomalies in medical scans



Autonomous Vehicles

Recognizing lanes, pedestrians, and obstacles



Retail

Cashier-less checkout and inventory tracking



Security

Facial recognition and surveillance analytics



A GROWING FIELD

billions

of images and video frames are processed by computer vision systems every single day, across phones, cars, and cameras worldwide.

From unlocking your phone to guiding surgical robots — vision is everywhere.

Challenges & The Road Ahead



CURRENT CHALLENGES

- Requires massive labeled datasets to train reliably
- Struggles with unusual lighting, angles, or occlusion
- Can inherit bias present in training data
- Demands heavy computing power for real-time use



WHERE IT'S HEADED

- Multimodal models combining vision with language
- Faster, smaller models running directly on devices
- Better generalization with less labeled data
- Tighter integration with robotics and real-time decision-making

THANK
YOU